

MovingAveragePerChannelMinMaxObserver

<https://pytorch.org/docs/stable/generated/torch.ao.quantization.observer.MovingAveragePerChannelMinMaxObserver.html>

Description:

Observer module for computing the quantization parameters based on the running per channel min and max values. This observer uses the tensor min/max statistics to compute the per channel quantization parameters. The module records the running minimum and maximum of incoming tensors, and uses this statistic to compute the quantization parameters. `averaging_constant` – Averaging constant for min/max. `ch_axis` – Channel axis `dtype` – Quantized data type

Function Signature

```
class
torch.ao.quantization.observer.MovingAveragePerChannelMinMax
Observer(averaging_constant=0.01, ch_axis=0,
dtype=torch.quint8, qscheme=torch.per_channel_affine,
reduce_range=False, quant_min=None, quant_max=None,
eps=1.1920928955078125e-07, is_dynamic=False, **kwargs)
[source]#
```

Parameters

Notes & Warnings

NOTE

Note If the running minimum equals to the running maximum, the scales and zero_points are set to 1.0 and 0.

Detailed Documentation

Documentation

Observer module for computing the quantization parameters based on the running per channel min and max values.

This observer uses the tensor min/max statistics to compute the per channel quantization parameters. The module records the running minimum and maximum of incoming tensors, and uses this statistic to compute the quantization parameters.

averaging_constant – Averaging constant for min/max.

ch_axis – Channel axis

dtype – Quantized data type

qscheme – Quantization scheme to be used

reduce_range – Reduces the range of the quantized data type by 1 bit

quant_min – Minimum quantization value. If unspecified, it will follow the 8-bit setup.

`quant_max` – Maximum quantization value. If unspecified, it will follow the 8-bit setup.

`eps (Tensor)` – Epsilon value for float32, Defaults to `torch.finfo(torch.float32).eps`.

The quantization parameters are computed the same way as in `MovingAverageMinMaxObserver`, with the difference that the running min/max values are stored per channel. Scales and zero points are thus computed per channel as well.

If the running minimum equals to the running maximum, the scales and `zero_points` are set to 1.0 and 0.